

Gain Peaking & Overshoot

Assumption: 2-active Poles
2-interacting Poles

$$\text{ex. } a(jf) = a_0 / (1+jf/f_1)(1+jf/f_2)(1+jf/f_3) \quad \text{neglect } f_3 \gg f_1, f_2$$

assumption: $\beta(jf) = \beta_0$

$$T(jf) = a_0 \beta_0 / (1+jf/f_1)(1+jf/f_2)$$

$$A(jf) = A_{ideal} / (1 + 1/T(jf))$$

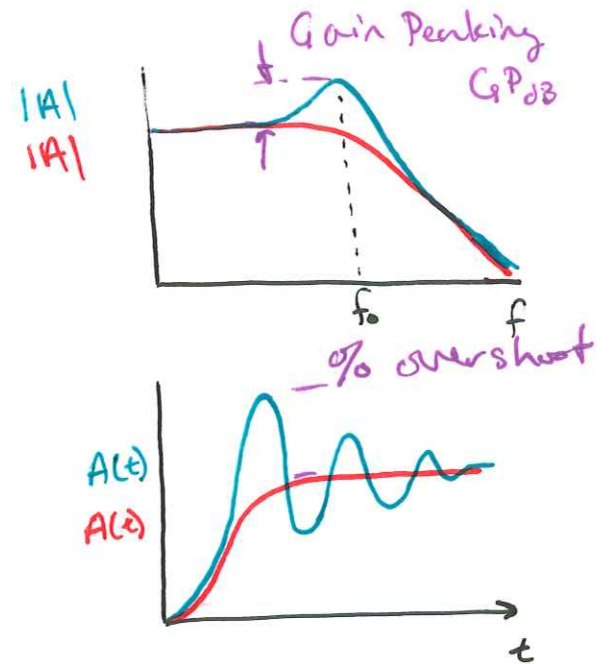
$$A(jf) = \frac{A_0}{1 + jf/Qf_0 - f^2/f_0^2}$$

if $f \ll f_0$ $A(jf) \approx A_0$

if $f \approx f_0$ $A(jf) \approx -jA_0Q \approx |A(jf)|_{max}$

$$GP_{dB} = 20 \log_{10} \left[\frac{2Q^2}{\sqrt{4Q^2 - 1}} \right] \quad \text{only if } Q > \frac{1}{\sqrt{2}}$$

if Q large, $GP_{dB} \approx 20 \log_{10} Q$



$$f_0^2 = a_0 \beta_0 f_1 f_2$$

$$Q = f_0 / [f_1 + f_2] \rightarrow \text{"Quality factor"}$$

related to damping factor, $\xi = 1/2Q$

if $Q > \frac{1}{\sqrt{2}} \rightarrow$ gain peaking occurs.

% overshoot — %OS

$$\%OS = 100\% \cdot e^{-\pi \xi / \sqrt{1-\xi^2}} \quad \xi = 1/2Q$$

$$\phi_m = \cos^{-1}(\sqrt{4\xi^4 + 1} - 2\xi^2) = \cos^{-1}(\sqrt{1 + 1/4Q^4} - 1/2Q^2)$$

Reminder: only 2 poles!

→ if more than 2: simulate!
(or use a solver)

Rough (quick) estimates of ϕ_m & GM

$$\phi_m = 180^\circ + \angle T(jf_x) \quad f_x \text{ is } |T(jf_x)| = 1 = 0\text{dB}$$

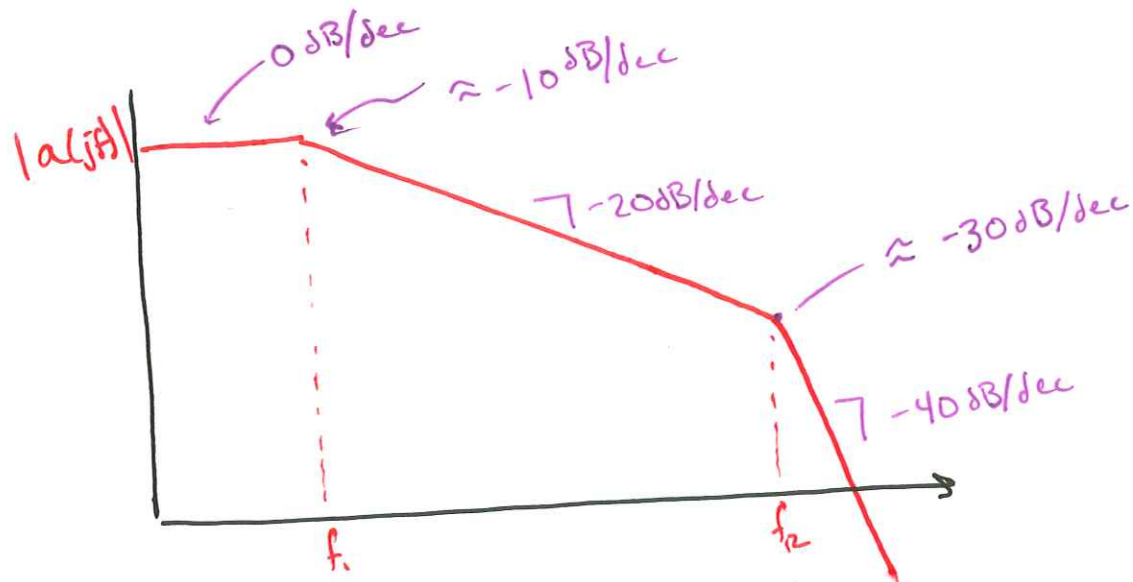
$$|T(jf_x)| = |a(jf_x)| | \beta(jf_x) | = 1$$

$$|a(jf_x)| = 1 / | \beta(jf_x) |$$

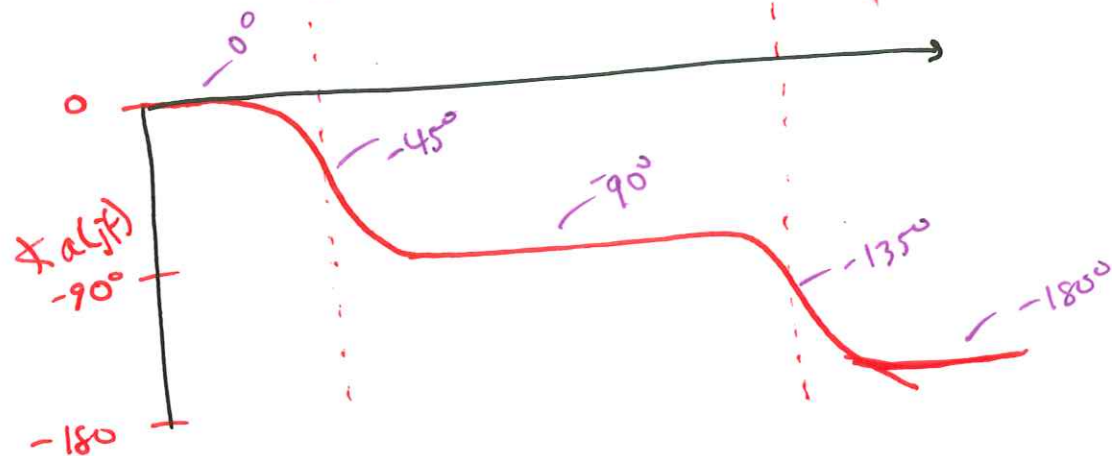
$$\phi_m = 180^\circ + \angle a(jf_x) + \angle \beta(jf_x) = 180^\circ + \angle a(jf_x) - \angle 1/\beta(jf_x)$$

with τ "unknown" but poles & zeros in $a(jf)$, $\beta(jf)$ known
AND poles are distinguishable (known & separate)
then ϕ_m can be estimated & easily.

Characterizations of $a(j\omega)$ (assume 2 poles)



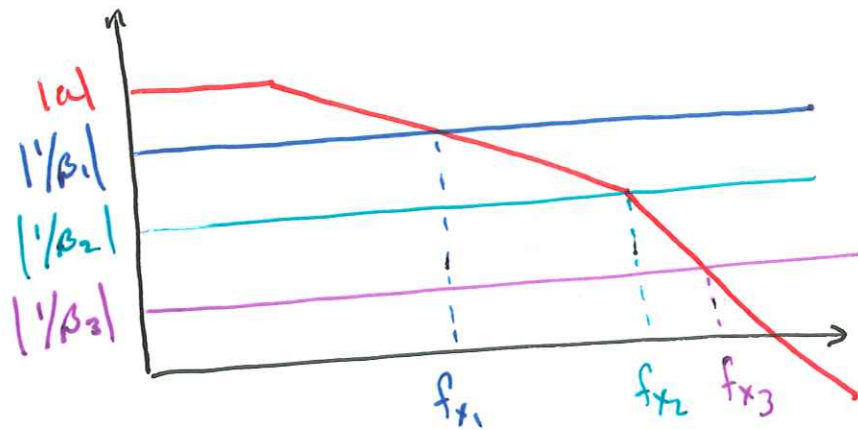
$$\angle a \approx 4.5 \cdot (\text{slope of } a \text{ in dB/dec})$$



Can do same thing with β (or $1/\beta$)

Ex. a is 2 pole
 β is real & constant (~~phase~~ phase = 0)

Rate-of-Closure
 Analysis.

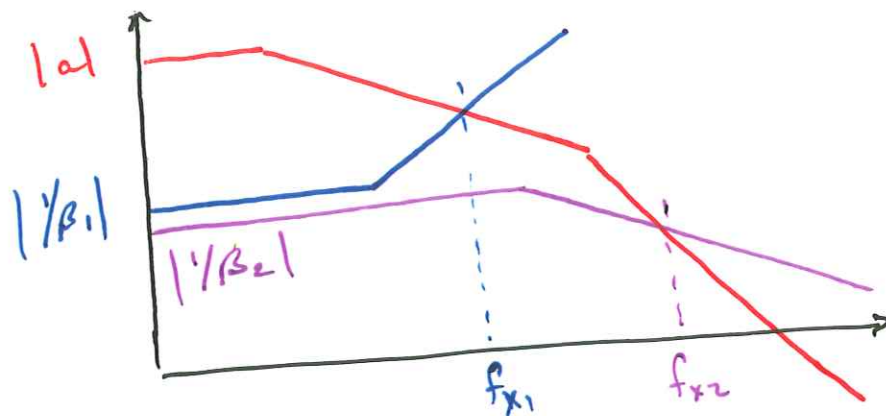


β_1 : at f_{x1} , a has -20 dB/dec , $1/\beta$ is 0 dB/dec
 $\phi_m = 180^\circ + \angle a \approx 180^\circ - 90^\circ = 90^\circ$

β_2 : f_{x2} a : -30 dB/dec ,
 $\phi_m \approx 180^\circ - 135^\circ = 45^\circ$

β_3 : f_{x3} a : -40 dB/dec
 $\phi_m \approx 180^\circ - 180^\circ = 0^\circ$ unstable!

Ex a has 2 pole, β is not constant.
can have zeros or poles

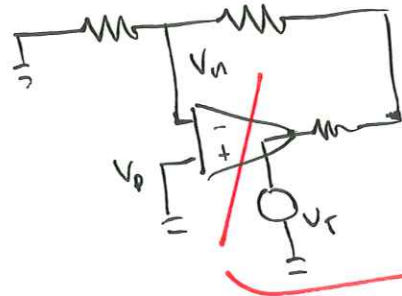


$\beta_1: f_{x1}$ $a: -20 \text{ dB/dec}$ $1/\beta: +20 \text{ dB/dec}$
 $a: -90^\circ$ $1/\beta: +90^\circ$
 $\phi_m = 180^\circ - 90^\circ - +90^\circ = 0^\circ$ unstable!

$\beta_2: f_{x2}$ $a: -40 \text{ dB/dec}$ $1/\beta: -20 \text{ dB/dec}$
 $a: -180^\circ$ $1/\beta: -90^\circ$
 $\phi_m = 180^\circ - 180^\circ - -90^\circ = 90^\circ$

Measuring return ratio

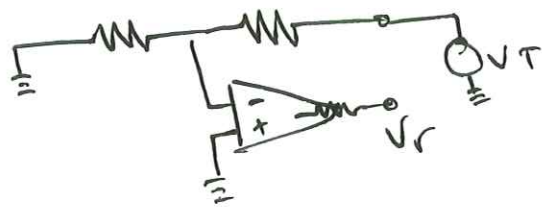
calculating:



calculate V_d with respect to
inserted voltage V_r

loop broken here.

Loop break ~~can~~ can occur anywhere in loop!

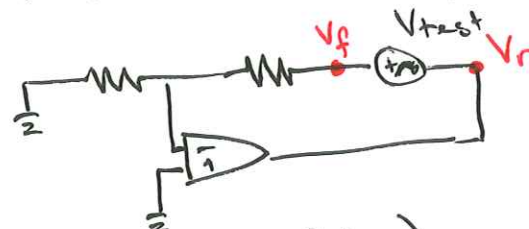


"effectively" doing same thing,
but at different location!
(not fully correct because of current considerations)

→ Do in situ

Middlebrook Technique

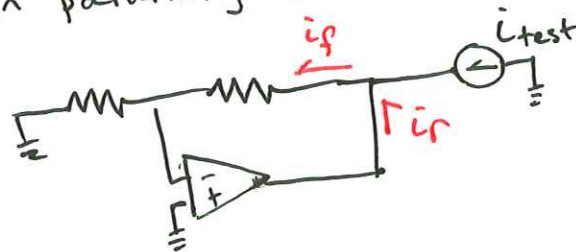
- 1) Set sources to zero
- 2) Cut feedback at desired point.
- 3) Add a floating ~~test~~ voltage v between these points



- 4) Measure V_f (voltage in forward direction of loop) and V_r (reverse direction) adjacent to the ~~supp~~ source.

$$T_V(jf) = -V_r/V_f$$

- 5) Reconnect cut (remove test voltage)
- 6) Add a current source [in parallel] at same cut location.



- 7) Measure i_f & i_r

$$T_I(jf) = -i_r/i_f$$

- 8) Calculate T : $\frac{1}{1+T} = \frac{1}{1+T_V} + \frac{1}{1+T_I} \rightarrow T = \frac{T_V T_I - 1}{T_V + T_I + 2}$

Consider $\frac{1}{1+T} = \frac{1}{1+T_V} + \frac{1}{1+T_I}$

Compare to $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

if $R_1 \ll R_2$, then $R \approx R_1$

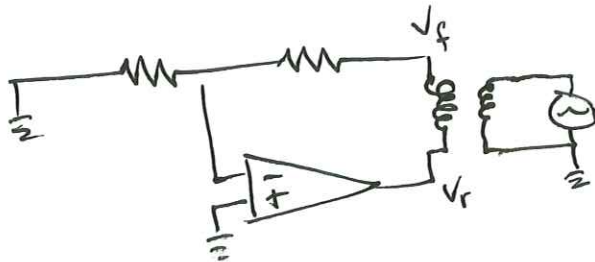
if $T_V \ll T_I$: $T \approx T_V$

if $T_V \gg T_I$: $T \approx T_I$

will depend on where placed in loop!

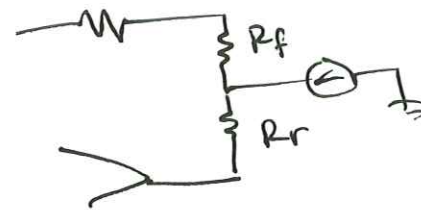
physically injecting floating AC voltage.

Use a transformer:



Measuring current:

Dummy resistors!



$R_f, R_r \ll R_{\text{feedback}}$